

Tianhao (Andy) Qin

Tel: (347) 247-8167 | andyqin918@gmail.com | LinkedIn: [linkedin.com/in/tianhao-qin](https://www.linkedin.com/in/tianhao-qin)

EDUCATION

Worcester Polytechnic Institute <i>Ph.D. in Robotics Engineering, Cumulative GPA: 4.00</i>	Worcester, MA Aug 2025 – Present
University of Michigan, College of Engineering <i>Master of Science in Robotics, Cumulative GPA: 4.00</i>	Ann Arbor, MI Aug 2023 – May 2025
New York University, Tandon School of Engineering <i>Bachelor of Science in Mechanical Engineering, Cumulative GPA: 3.89</i> Minors in Computer Science and Robotics	Brooklyn, NY Sep 2019 – May 2023

PUBLICATIONS

- T. Qin**, C. McCann, B. Calli, and J. Xiao, “Gripper-Aware Automatic Dense Packing of Irregular Objects,” in *International Symposium of Robotics Research (ISRR)*, 2026.
- G. Ma*, **T. Qin***, J. Liu, and M. Draelos, “OCT-DeformNet: Optical Coherence Tomography Guided Biological Tissue Shape Prediction for Robot Palpation in Microsurgery,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- T. Qin**, M. Draelos, and G. Formosa, “From Theory to Practice: A Hands-On Laboratory Curriculum for Graduate Medical Robotics Education,” in *ASEE Annual Conference & Exposition*, 2026.

EXPERIENCE

Worcester Polytechnic Institute, Adaptive and Intelligent Robotics Lab <i>Graduate Research Assistant Advisor: Prof. Jing Xiao</i> - Developed a gripper-aware planning algorithm that densely packs irregular objects with soft fin-ray grippers. - Deployed on a Franka Emika Panda, achieving 96% space utilization across 7 customized objects and 10 YCB objects.	Worcester, MA Aug 2025 – Present
University of Michigan, Image-Guided Medical Robotics Lab <i>Graduate Research Assistant PI: Prof. Mark Draelos</i> - Co-developed OCT-DeformNet, a deep learning model to predict tissue deformation under robot palpation with OCT data. - Built the synthetic data simulator, trained the model in PyTorch, and validated on UR5e robot. - Achieved average prediction error of 0.15mm on phantom model, 0.55mm/0.49mm on ex vivo chicken/porcine tissue.	Ann Arbor, MI Sep 2024 – Aug 2025
University of Michigan, Robotics Department <i>Graduate Student Instructor ROB 599 Medical Robotics</i> - Developed lab material including a 1-DoF bilateral teleoperation testbed and virtual fixtures on a 6-DoF haptic device. - Published the curriculum as first author at ASEE 2026.	Ann Arbor, MI Dec 2024 – May 2025
New York University, Computer Science and Engineering Department <i>Course Assistant CS 1114 Introduction to Programming and Problem Solving</i> Taught weekly 3-hour Python lab sections, graded assignments, and held 10 hrs/week office hours.	Brooklyn, NY Jan 2022 – May 2023
New York University, Future Labs <i>Product Engineer Intern “Trashy Wheels”</i> - Produced skateboarding wheels with biodegradable polyurethanes, earning an NYU Prototyping Fund award. - Designed injection molds with SOLIDWORKS and manufactured molds from polyethylene plastic blocks through CNC.	Brooklyn, NY Sep 2021 – Jan 2022

LEADERSHIP

NYU Robotic Design Team, Vertically Integrated Projects <i>Team Lead (2021-2023), CAD Engineer (2019-2021) NASA Robotic Mining Competition: Lunabotics</i> - Led a 50-member team to design and build a lunar excavation rover for icy regolith. - Developed ConOp, System Requirements, and Verification Plans and implemented Agile project management.	Brooklyn, NY Sep 2019 – May 2023
---	-------------------------------------

SKILLS & INTERESTS

- | | |
|--|--|
| <ul style="list-style-type: none">Languages: Python, C++, MATLABRobotics: ROS2, PyBullet, IsaacLab, Gazebo, PyTorch | <ul style="list-style-type: none">CAD/Fab: SOLIDWORKS, ANSYS, OnShapeResearch areas: Manipulation, Robot Learning, Planning |
|--|--|